

EXAMPLE 6.14

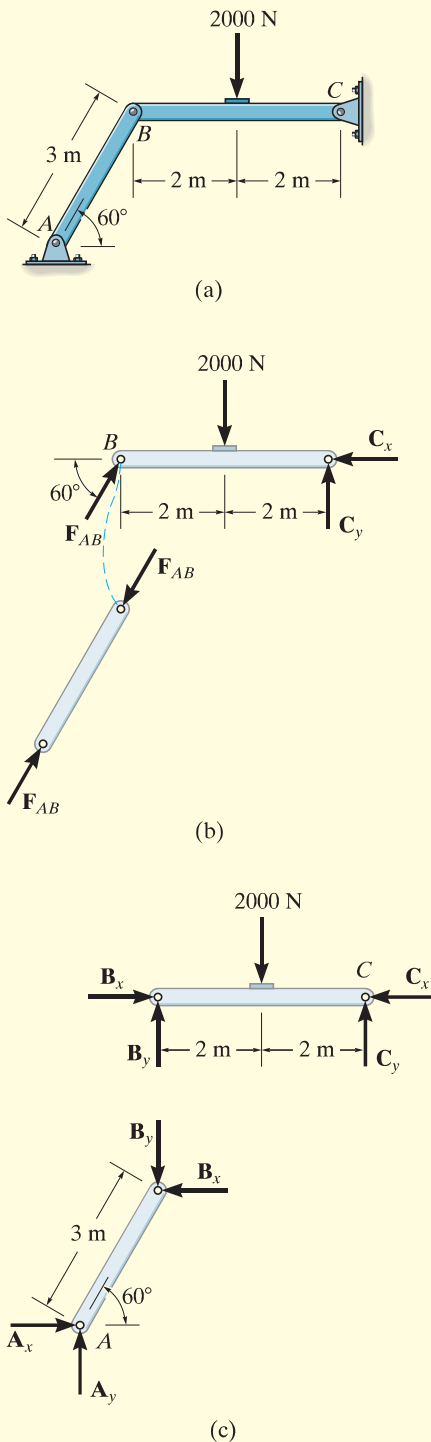


Fig. 6-26

Determine the horizontal and vertical components of force which the pin at C exerts on member BC of the frame in Fig. 6-26a.

SOLUTION I

Free-Body Diagrams. By inspection it can be seen that AB is a two-force member. The free-body diagrams are shown in Fig. 6-26b.

Equations of Equilibrium. The *three unknowns* can be determined by applying the three equations of equilibrium to member CB .

$$\zeta + \sum M_C = 0; 2000 \text{ N}(2 \text{ m}) - (F_{AB} \sin 60^\circ)(4 \text{ m}) = 0 \quad F_{AB} = 1154.7 \text{ N}$$

$$\rightarrow \sum F_x = 0; 1154.7 \cos 60^\circ \text{ N} - C_x = 0 \quad C_x = 577 \text{ N} \quad \text{Ans.}$$

$$+\uparrow \sum F_y = 0; 1154.7 \sin 60^\circ \text{ N} - 2000 \text{ N} + C_y = 0 \quad C_y = 1000 \text{ N} \quad \text{Ans.}$$

SOLUTION II

Free-Body Diagrams. If one does not recognize that AB is a two-force member, then more work is involved in solving this problem. The free-body diagrams are shown in Fig. 6-26c.

Equations of Equilibrium. The *six unknowns* are determined by applying the three equations of equilibrium to each member.

Member AB

$$\zeta + \sum M_A = 0; B_x(3 \sin 60^\circ \text{ m}) - B_y(3 \cos 60^\circ \text{ m}) = 0 \quad (1)$$

$$\rightarrow \sum F_x = 0; A_x - B_x = 0 \quad (2)$$

$$+\uparrow \sum F_y = 0; A_y - B_y = 0 \quad (3)$$

Member BC

$$\zeta + \sum M_C = 0; 2000 \text{ N}(2 \text{ m}) - B_y(4 \text{ m}) = 0 \quad (4)$$

$$\rightarrow \sum F_x = 0; B_x - C_x = 0 \quad (5)$$

$$+\uparrow \sum F_y = 0; B_y - 2000 \text{ N} + C_y = 0 \quad (6)$$

The results for C_x and C_y can be determined by solving these equations in the following sequence: 4, 1, 5, then 6. The results are

$$B_y = 1000 \text{ N}$$

$$B_x = 577 \text{ N}$$

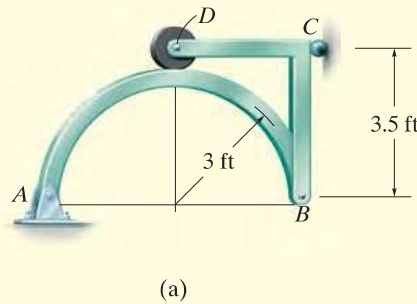
$$C_x = 577 \text{ N} \quad \text{Ans.}$$

$$C_y = 1000 \text{ N} \quad \text{Ans.}$$

By comparison, Solution I is simpler since the requirement that F_{AB} in Fig. 6-26b be equal, opposite, and collinear at the ends of member AB automatically satisfies Eqs. 1, 2, and 3 above and therefore eliminates the need to write these equations. *As a result, save yourself some time and effort by always identifying the two-force members before starting the analysis!*

EXAMPLE 6.17

The smooth disk shown in Fig. 6–29a is pinned at D and has a weight of 20 lb. Neglecting the weights of the other members, determine the horizontal and vertical components of reaction at pins B and D .

**SOLUTION**

Free-Body Diagrams. The free-body diagrams of the entire frame and each of its members are shown in Fig. 6–29b.

Equations of Equilibrium. The eight unknowns can of course be obtained by applying the eight equilibrium equations to each member—three to member AB , three to member BCD , and two to the disk. (Moment equilibrium is automatically satisfied for the disk.) If this is done, however, all the results can be obtained only from a simultaneous solution of some of the equations. (Try it and find out.) To avoid this situation, it is best first to determine the three support reactions on the *entire* frame; then, using these results, the remaining five equilibrium equations can be applied to two other parts in order to solve successively for the other unknowns.

Entire Frame

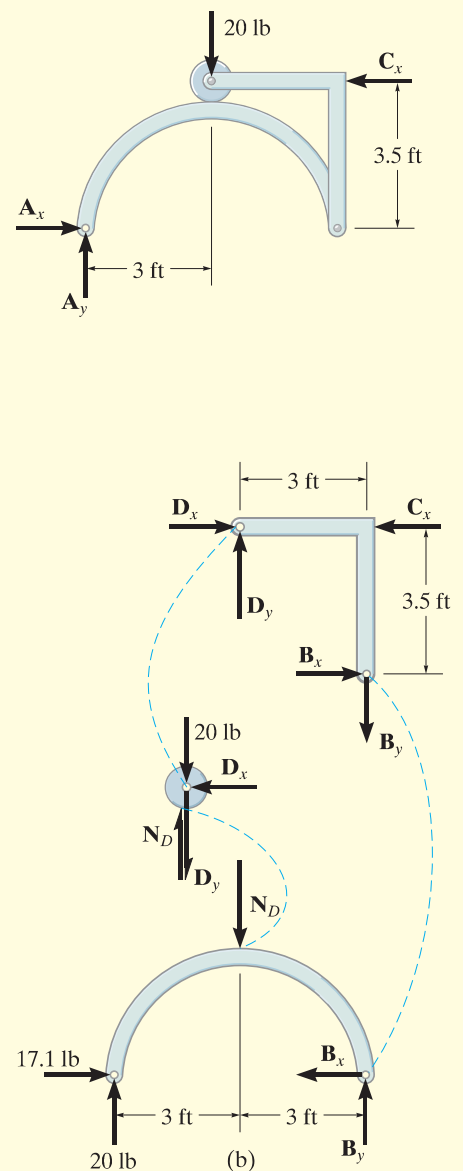
$$\begin{aligned} \zeta + \sum M_A = 0; & \quad -20 \text{ lb} (3 \text{ ft}) + C_x (3.5 \text{ ft}) = 0 & C_x = 17.1 \text{ lb} \\ \rightarrow \sum F_x = 0; & \quad A_x - 17.1 \text{ lb} = 0 & A_x = 17.1 \text{ lb} \\ + \uparrow \sum F_y = 0; & \quad A_y - 20 \text{ lb} = 0 & A_y = 20 \text{ lb} \end{aligned}$$

Member AB

$$\begin{aligned} \rightarrow \sum F_x = 0; & \quad 17.1 \text{ lb} - B_x = 0 & B_x = 17.1 \text{ lb} & \text{Ans.} \\ \zeta + \sum M_B = 0; & \quad -20 \text{ lb} (6 \text{ ft}) + N_D (3 \text{ ft}) = 0 & N_D = 40 \text{ lb} \\ + \uparrow \sum F_y = 0; & \quad 20 \text{ lb} - 40 \text{ lb} + B_y = 0 & B_y = 20 \text{ lb} & \text{Ans.} \end{aligned}$$

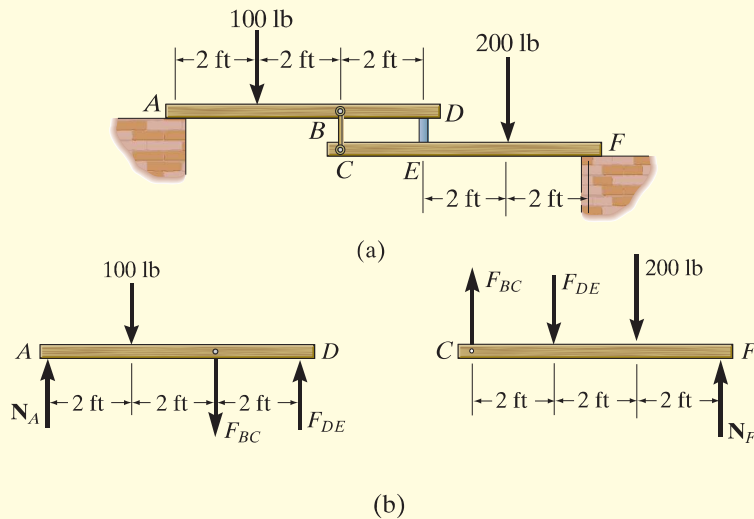
Disk

$$\begin{aligned} \rightarrow \sum F_x = 0; & \quad D_x = 0 & \text{Ans.} \\ + \uparrow \sum F_y = 0; & \quad 40 \text{ lb} - 20 \text{ lb} - D_y = 0 & D_y = 20 \text{ lb} & \text{Ans.} \end{aligned}$$

**Fig. 6–29**

EXAMPLE 6.19

The two planks in Fig. 6–31*a* are connected together by cable *BC* and a smooth spacer *DE*. Determine the reactions at the smooth supports *A* and *F*, and also find the force developed in the cable and spacer.

**Fig. 6–31****SOLUTION**

Free-Body Diagrams. The free-body diagram of each plank is shown in Fig. 6–31*b*. It is important to apply Newton's third law to the interaction forces as shown.

Equations of Equilibrium. For plank *AD*,

$$\zeta + \Sigma M_A = 0; \quad F_{DE}(6 \text{ ft}) - F_{BC}(4 \text{ ft}) - 100 \text{ lb}(2 \text{ ft}) = 0$$

For plank *CF*,

$$\zeta + \Sigma M_F = 0; \quad F_{DE}(4 \text{ ft}) - F_{BC}(6 \text{ ft}) + 200 \text{ lb}(2 \text{ ft}) = 0$$

Solving simultaneously,

$$F_{DE} = 140 \text{ lb} \quad F_{BC} = 160 \text{ lb} \quad \text{Ans.}$$

Using these results, for plank *AD*,

$$+\uparrow \Sigma F_y = 0; \quad N_A + 140 \text{ lb} - 160 \text{ lb} - 100 \text{ lb} = 0$$

$$N_A = 120 \text{ lb} \quad \text{Ans.}$$

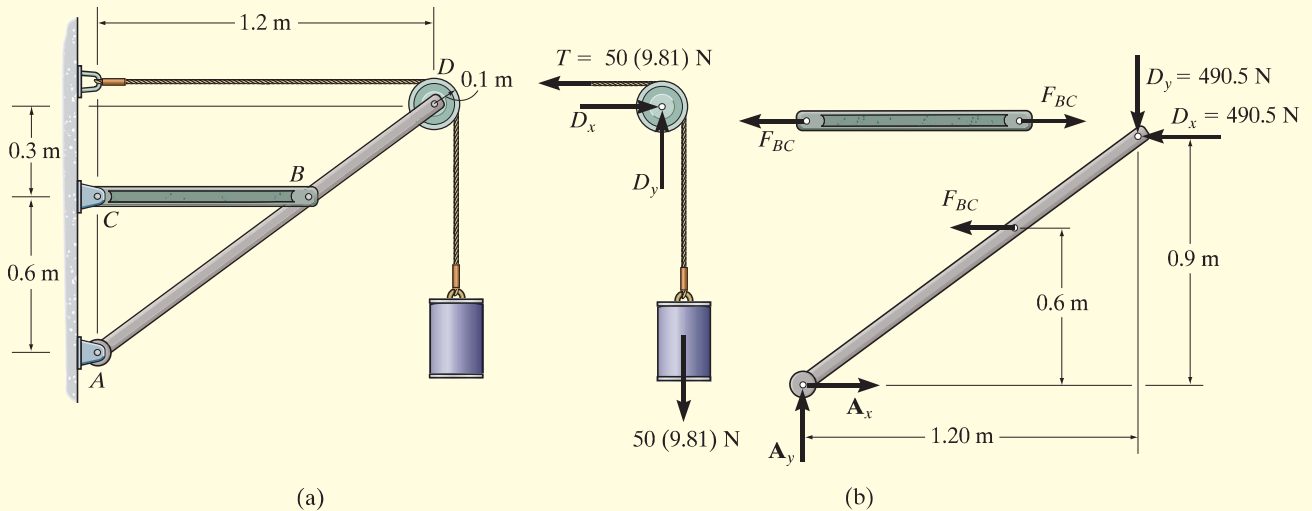
And for plank *CF*,

$$+\uparrow \Sigma F_y = 0; \quad N_F + 160 \text{ lb} - 140 \text{ lb} - 200 \text{ lb} = 0$$

$$N_F = 180 \text{ lb} \quad \text{Ans.}$$

EXAMPLE 6.21

The frame in Fig. 6–33*a* supports the 50-kg cylinder. Determine the horizontal and vertical components of reaction at *A* and the force at *C*.

**Fig. 6–33****SOLUTION**

Free-Body Diagrams. The free-body diagram of pulley *D*, along with the cylinder and a portion of the cord (a system), is shown in Fig. 6–33*b*. Member *BC* is a two-force member as indicated by its free-body diagram. The free-body diagram of member *ABD* is also shown.

Equations of Equilibrium. We will begin by analyzing the equilibrium of the pulley. The moment equation of equilibrium is automatically satisfied with $T = 50(9.81)$ N, and so

$$\begin{aligned} \rightarrow \Sigma F_x = 0; \quad D_x - 50(9.81) \text{ N} &= 0 & D_x &= 490.5 \text{ N} \\ + \uparrow \Sigma F_y = 0; \quad D_y - 50(9.81) \text{ N} &= 0 & D_y &= 490.5 \text{ N} \end{aligned} \quad \text{Ans.}$$

Using these results, F_{BC} can be determined by summing moments about point *A* on member *ABD*.

$$\begin{aligned} \zeta + \Sigma M_A = 0; \quad F_{BC}(0.6 \text{ m}) + 490.5 \text{ N}(0.9 \text{ m}) - 490.5 \text{ N}(1.20 \text{ m}) &= 0 \\ F_{BC} &= 245.25 \text{ N} \end{aligned} \quad \text{Ans.}$$

Now A_x and A_y can be determined by summing forces.

$$\begin{aligned} \rightarrow \Sigma F_x = 0; \quad A_x - 245.25 \text{ N} - 490.5 \text{ N} &= 0 & A_x &= 736 \text{ N} \\ + \uparrow \Sigma F_y = 0; \quad A_y - 490.5 \text{ N} &= 0 & A_y &= 490.5 \text{ N} \end{aligned} \quad \text{Ans.}$$